

The Process of Science:

The Scientific Method

Mr. Queenan – Biology 9

Unit 1.3

For more information, see text Ch 1.1 (p 11-16) & Ch 1.2 (p 17-21)

Questions?

- Essential Question: Can we have knowledge without order?
- What are examples of ways we seek to make order of our world?
 - Religion, Government, Ethics, Science
- What makes science different?

The Process of Science

- Biology, and even more broadly science, is a broad subject, with many different disciplines
- How is it possible for all scientists across all disciplines to speak a common language for gaining new knowledge?
- **The Scientific Method**
 - Science is based upon inquiry, the search for information and explanations of natural phenomena.
 - Scientific Method is a standard series of steps used in gathering and gaining new knowledge

The Scientific Method

- Steps
 1. Make an Observation
 2. Ask a Question
 3. Make a Hypothesis
 4. Perform an Experiment(s)
 5. Collect and Analyze Data
 6. Make Conclusions

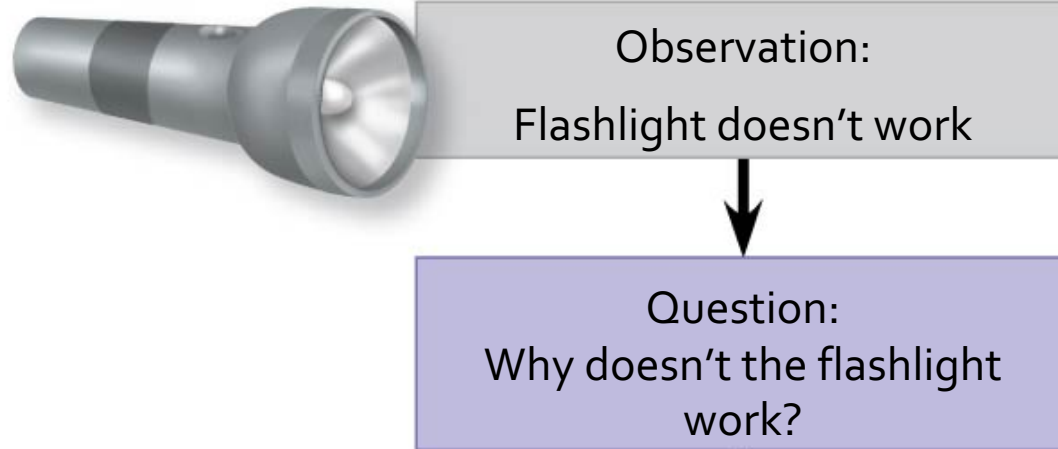
The Scientific Method

- **Observations and Questions**
 - What are the ways that we can make observations of **phenomena**, or a natural event?
 - Use of our Senses
 - Use of Tools & Technology
 - Qualitative (descriptive) vs Quantitative (numerical) Data
- What makes a question good?
- What makes a question good in the field of science?

The Scientific Method

- General Case Study

Your flashlight doesn't work



The Scientific Method

- **Hypothesis**

- A hypothesis is a possible explanation for a natural event
- Scientists use **inductive reasoning**, or creative thinking to combine isolated facts into a cohesive whole
 - Inductive reasoning makes generalizations based on collecting and analyzing a large number of specific observations.
- A hypothesis is not merely a guess.
 - Informed statement that can be tested according to the process of science!

The Scientific Method

- Experimentation
 - Scientist carries out an **experiment**, or series of procedures
- Use **Deductive Reasoning**
 - Make a prediction based on knowledge of the factors in the experiment
 - Ex. for flashlight case study
 - Predict 2 reasons why flashlight doesn't work

The Scientific Method

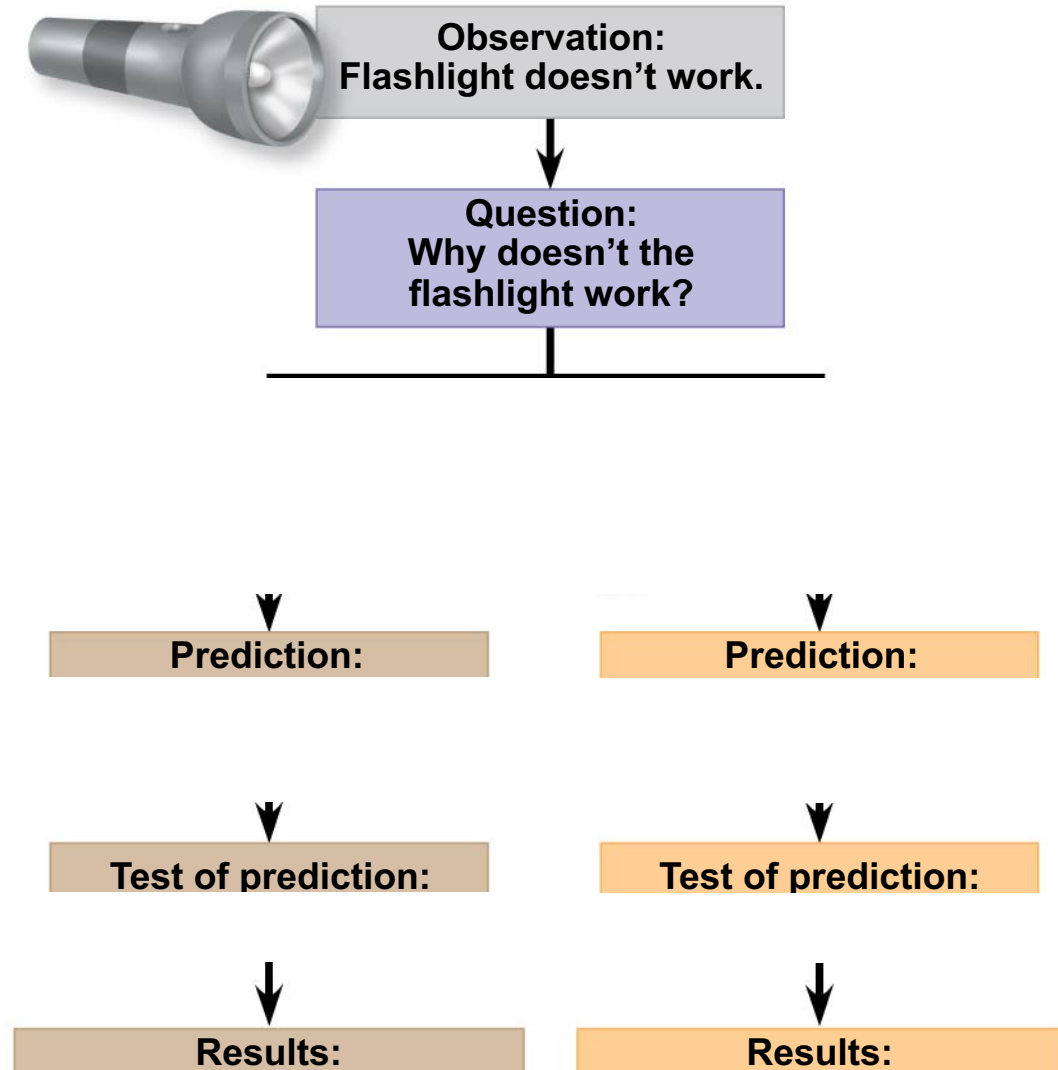
- Experimental Design
 - Good design is important to ensure that results observed are a result of experimental procedure
- **Experimental Variable**
 - A factor in an experiment that is changed
- **Control**
 - Unchanging factors in an experiment
- Test group and control group
 - Test groups exposed to experimental variable while control group is not.
- If results show no difference between control and test groups, hypothesis not supported
- If results show difference in relation to predictions, hypothesis supported
- If results show difference not in relation to prediction, different hypothesis needed or experiment needs better controls

The Scientific Method

- Results
 - Scientists will either directly observe or will model using technology to collect **data**.
 - Data is any result of an experiment, and can be observational or numerical
 - Statistical Analysis is needed to determine if data results are *statistically significant*
- Conclusions
 - After analyzing results, determine if hypothesis is supported or not
 - Science is never ending; can lead to new questions
 - Findings reported to scientific community through reports, journals, presentations, etc.
 - Results must be replicable by scientific community

The Scientific Method

General
Case
Study
Continued



The Process of Science

- A scientific **theory** is
 - much broader in scope than a hypothesis and
 - supported by a large and usually growing body of evidence.
- Examples of 2 Fundamental Theories of Biology
 - Evolution – All living things have a common ancestor, but each is adapted to a particular way of life
 - Cell – All organisms are composed of cells, and new cells only come from pre-existing cells
- When a theory is supported by all experimentation over a longer period of time, it may be referred to as a **Principle** or **Law**
- Science is a social activity in which scientists
 - work in teams,
 - share information through peer-reviewed publications, meetings, and personal communication, and
 - build on and confirm each other's work.

The Scientific Method

Case Study 2

- Scientists conducted a **controlled experiment** to test the hypothesis that color patterns have evolved as adaptations that protect animals from predation.
- The experiment compared an experimental group consisting of noncamouflaged mice models and a control group consisting of camouflaged models that matched the mice native to each area.
- The groups differed by only one factor, the coloration of the mouse models.

The Scientific Method

Case Study 2

Beach population



Inland population



TABLE 1.9 | RESULTS FROM CAMOUFLAGE EXPERIMENT

Habitat	Number of Attacks		% Attacks on Non-camouflaged Models
	On Camouflaged Models	On Non-camouflaged Models	
Beach (light habitat)	2	5	71%
Inland (dark habitat)	5	16	76%

Data from S. N. Vignieri et al., The selective advantage of crypsis in mice, *Evolution* 64: 2153–2158 (2010).

- The noncamouflaged models had a much higher percentage of attacks in the beach and inland habitats and
- These data fit the key prediction of the camouflage hypothesis.

The Scientific Method

Case Study 2

Case Study In Class

- You notice that you left some skittles out on your desk when you went to bed last night. The skittles were next to your glass of water, which had created a puddle. You notice that the color and some of the inside of the skittle has dissolved into the puddle.
- Craft a line of scientific inquiry based on this observation, using the steps of the scientific method.
 1. Make an Observation
 2. Ask a Question
 3. Make a Hypothesis
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 5. Collect and Analyze Data
 6. Make Conclusions